

Exotic physics in 2d materials: computational study of twisted homobilayers of the transition-metal dichalcogenide WSe_2

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Abstract. Twisted homobilayers of the transition metal dichalcogenide WSe_2 are believed to be well-described by the moiré Hubbard model. Using the density matrix renormalization group algorithm, we investigate an extended moiré Hubbard model, finding a pronounced Mott insulating state at half-filling, and indications of other insulating states at 1/8 and 3/8 fillings, which we conjecture may be generalized Wigner crystals.

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Finally, I would like to thank J. C. Bridgeman and C. T. Chubb (authors of [1]) for generously open-sourcing the LaTeX code they used to generate beautiful tensor network diagrams, which I have modified and used throughout this work.

Declaration of authorship. I declare that all material in this thesis is my own work, except where there is clear acknowledgement and appropriate reference to the work of others.

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1. Introduction

In his now-famous 1972 article *More is different* [4], P. W. Anderson notes that a new understanding is dawning upon him and his colleagues within the field of many-body physics — the fundamental laws which govern the elementary particles cannot adequately describe the effective behaviour of meso- or macroscopic systems. He argues that we are approaching the death of the reductionist view that all scientific fields are simply the extrapolation of more fundamental laws.

Today, that death is well established. The modern landscape of condensed matter physics (which Anderson pioneered) is dominated by complex emergent phenomena: quasiparticle excitations such as the Majorana fermion [5], unconventional high-temperature superconductivity [6], and the fractional quantum Hall effect [7] to name just a small selection of current topics that draw heavy research attention. As a general rule, the complexity of the physics in many-body systems tends to increase as the interactions between the constituents become stronger [8].

1.1. Interacting many-body systems

On one end (and considering for the sake of concreteness that we are modelling valence electrons occupying orbitals of ions which form a lattice structure), we have the non-interacting limit in which electrons have no correlation with one another — they are permitted only to tunnel (or 'hop' as is commonly used in the literature) to neighbouring atoms in the lattice, mathematically described by a coupling constant t. Such a model has no interactions between constituent electrons, and so we can effectively construct the space of many-body wavefunctions as a tensor product of single-electron states, which commonly have well-understood analytical solutions.

Of course, no real particle is non-interacting, and so this might seem like a naive approach. On the contrary, free electron models work remarkably well, precisely because of emergent behaviour. Electrons in metals (a form of matter in which we intuitively expect electrons to strongly repel one another) are well-approximated as a non-interacting Fermi gas; the interacting Fermi liquid theory explains this by noting a one-to-one correspondence between its interacting electron quasiparticles (which differ from electrons

through a renormalized mass) and the non-interacting electrons of the Fermi gas [9].

The non-interacting Fermi gas does of course fail in a number of basic ways, perhaps most notably by completely failing to predict the existence of insulators, semiconductors, superconductors, or semimetals [10]. The addition of ion potentials (the so-called nearly-free electron model) enables the calculation of band structures. Weakly-interacting models can be derived through perturbation theory — to grossly oversimplify, we choose a non-interacting model which resembles the interacting model we are interested in, and then add corrections to the non-interacting theory to build an interacting one. Perturbation theory is a mature and powerful method, but as the name implies it relies on the assumption that our interactions are only small perturbations; when the interaction strength U is of comparable magnitude to the hopping constant t, we enter the regime of *strongly correlated physics*.

1.2. Strongly correlated materials

Since its independent proposal by Hubbard [11], Gutzwiller [12], and Kanamori [13] in 1963, the Hubbard model has become paradigmatic in the study of strong interactions in many-body systems. Written in the notation of second quantization, the general form of the Hubbard Hamiltonian $H = H_0 + H_I$ is the sum of a kinetic term H_0 and an on-site density interaction term H_I [14],

$$\begin{aligned} H_0 &= -\sum_{i,j} \sum_{\sigma=\uparrow,\downarrow} \left(t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.} \right), \\ H_I &= U \sum_i n_{i\uparrow} n_{i\downarrow}, \end{aligned} \tag{1}$$

where t_{ij} is the hopping constant between lattice sites i and j, $c_{i\sigma}$ ($c_{i\sigma}^\dagger$) is the fermionic annihilation (creation) operator, h.c. is an abbreviation for 'hermitian conjugate' (it is enforced that the Hamiltonian operator must be Hermitian), U is the interaction strength, and $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$ is the occupation number operator for spin σ on site i. Typically, $t_{ij} > 0$, and in the usual case of considering spin-$\frac{1}{2}$ electrons on the lattice, $U < 0$ represents the repulsive Coulomb interaction. When we are in the regime such that $|U|$ is on the order of $|t|$, there exists no small parameter with respect to which we can expand the model as a perturbative series while preserving the essential physics of the model.

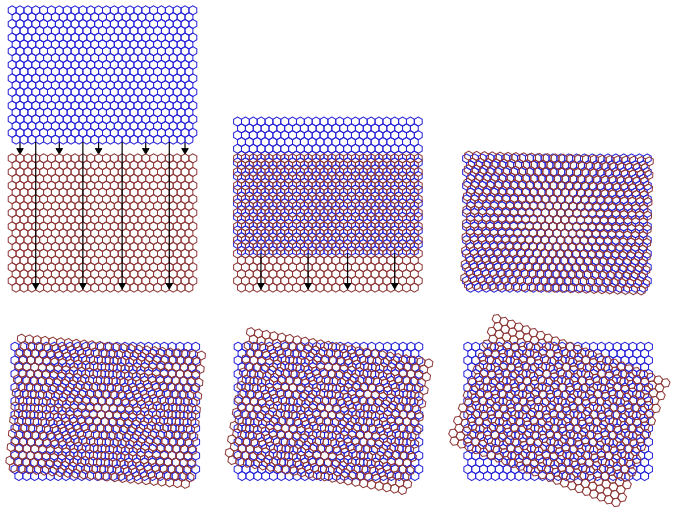


FIG. 1. The stacking and then twisting of 2d lattices creates moiré superlattices, significantly altering the properties of the system.

The Hubbard model is a so-called 'effective' model, in the sense that it represents only the essential characteristics of the physical microscopic system, discarding as many minor complications as is possible without damaging its ability to capture the underlying physics. One might then hope that these effective models are straightforward to solve; unfortunately, this is not the case. The Hubbard model has only been established to have an exact Bethe-ansatz solution in one dimension [15]. In two and three dimensions, numerical studies [16] have unearthed a rich phenomenology of phases and regimes, including unconventional superconductivity, antiferromagnetism, charge-density waves, generalized Wigner crystals, spin liquids, and more (see [14] and references therein).

Of the so-called strongly correlated materials (increasingly also becoming known as quantum materials [17]) which these models aim to emulate, the class of moiré materials has in recent years inspired intense theoretical, computational, and experimental research [18].

1.2.1. Moiré materials

In this work, we define 2d materials as those materials which are generally a few atoms thick and are well-described by (quasi)particle excitations confined to two dimensions (following the definition formulated by [19]). When 2d materials are stacked and either

- have a lattice constant mismatch,
- or a lattice constant mismatch is induced by introducing a relative twist angle between the layers

the resulting system (called a moiré material) has significantly altered properties compared to either of the original materials. When 2d materials are stacked, new properties appear only if the interlayer coupling is sufficiently strong, such that charge can redistribute itself among the layers; otherwise, the layers are essentially independent and demonstrate no new properties. Twisting of layers induces such an interlayer coupling. The moiré pattern which forms when periodic structures are misaligned (see Fig. 1) induces atomic reconstruction, meaning that the atoms shift position to a new equilibrium configuration which balances the inter- and intralayer coupling – effectively, this equilibrium configuration takes the form of the moiré pattern, and we are left with a moiré superlattice [19].

The superlattice constant a_M is significantly larger than the original atomic lattice constant a. This difference in length scale separates the low-energy physics (determined by a_M) and the high-energy physics (determined by a). In the low-energy regime (the limit where we can discard the influence of a and consider only a_M), the sites of the moiré superlattice are essentially artificial atoms, in the sense that the electrons in the system see a periodic potential [20]. Bloch's theorem then applies, and the bands which form are flat [21]. This length scale makes moiré materials uniquely suited to studies of strongly correlated physics [18, 22], because the kinetic energy E_k of electrons on the moiré superlattice scales as $E_k \propto a_M^{-2}$ [21], while the on-site Coulomb interaction U scales as $U \propto a_M^{-1/2}$ [23]. Therefore, U dominates E_k for large a_M, and the electrons become strongly correlated – since a_M is relatively straightforwardly tuned by altering the twist angle, moiré materials form an ideal platform for realizing the strong correlations regime of the Hubbard model.

1.2.2. Transition metal dichalcogenides

In this work, we will focus in particular on semiconductor moiré materials, for two primary reasons:

1. Unlike, for example, twisted bilayer graphene, semiconductor moiré materials feature the formation of flat bands continuously over twist angles [24], whereas twisted bilayer graphene requires specific

'magic' angles for flat band formation [25].

2. The flat bands in semiconductor moiré materials are well-described by local tight-binding models [24, 26], where we note that the Hubbard Hamiltonian (Eq. 1) is the sum of the tight-binding Hamiltonian with hopping t and an interaction term of strength U.

Further, we limit ourselves to tungsten diselenide (WSe_2), part of the broader family of transition-metal dichalcogenides (TMDs), in which considerable progress has been made in recent years regarding the experimental ground-state phase diagram [21]. TMDs, and in particular both twisted WSe_2 homobilayers [27] and angle-aligned WSe_2/WS_2 heterobilayers [28, 29], have been found to exhibit the behaviour of a Mott insulating state at half-filling of the lattice. At commensurate fractional fillings, charge-ordered states emerge in angle-aligned WSe_2/WS_2 heterobilayers, including generalized Wigner crystals (which preserve rotation symmetry) [29] and stripe phases (which spontaneously break rotation symmetry) [30].

Further yet, we limit ourselves to studies of the ground state of twisted WSe_2 homobilayers, in which the recent observation of a sharp drop in resistivity has been interpreted as an indication of a superconducting state [31]; we make this restriction as it greatly simplifies the task at hand, and we justify it by noting that (1) the ground state largely characterizes the low-energy physics of a system, and (2) the ground state provides a basis upon which excited states may be constructed in future studies.

Therefore, we can summarise our purpose as the following: we seek to numerically establish the phenomena present in the ground state of twisted WSe_2 homobilayers (abbreviated as tWSe_2 from this point onwards), as modelled by an extended variant of the triangular lattice Hubbard model called the moiré Hubbard model; we will introduce the model at a later point (Sec. 3).

1.3. Numerical many-body physics

Interacting many-body electron problems pose some of the greatest computational challenges in contemporary science [32]. For a system of N particles, each described by a single-particle Hilbert space $\mathcal{H}_i$, $i \in [1, ..., N]$, of dimension $\dim(\mathcal{H}_i) = d$, the total many-body Hilbert space is given by their tensor

product,

$$\mathcal{H} = \bigotimes_{i=1}^{N} \mathcal{H}_i$$

and has a dimension $\dim(\mathcal{H}) = d^N$ — the size of the many-body Hilbert space scales exponentially with the number of particles under consideration. A naive attempt to brute-force even a respectable approximation of the thermodynamic limit ($N \to \infty$) is therefore intractable, in that the size of the many-body wavefunction will rapidly outgrow the storage capabilities of any contemporary computer. Developing suitable methods which side-step the exponential growth of the many-body Hilbert space has become an integral component of modern condensed matter physics.

As it stands, all available generic methods can be categorised as either (1) methods which scale exponentially with N, but provide exact results, or (2) methods which scale linearly or polynomially with N, but provide approximations by construction [32]. Of the former, exact diagonalization of the Hamiltonian by Lancoz's method [33] is currently feasible for up to about $N = 50$ [8].

Of the latter, quantum Monte Carlo (QMC) methods have emerged as one of the most important family of methods in many-body simulations [34]. Exploiting the fact that a quantum partition function can be mapped to a summation of classical Boltzmann weights, QMC avoids performing computations on the entire many-body Hilbert space by employing standard Monte Carlo methods, which rely on system-size-agnostic stochastic sampling. Given sufficient computation time, QMC methods can reach arbitrary precision. Unfortunately, in the case of fermionic or geometrically frustrated systems, the quantum to classical mapping can suffer from what is called the 'sign problem' — the Boltzmann weights are not necessarily positive definite, and so can not be immediately interpreted as probabilities in the spirit of classical Monte Carlo (see [34, 35] for recent reviews of QMC and the sign problem). Cleverly constructed QMC methods which solve the sign problem for specific models exist, but it has been shown that the sign problem is NP-hard [36], and so there almost certainly does not exist a generic polynomial time solution.

The triangular lattice is the simplest lattice which exhibits geometrical frustration (see Fig. 2), and so QMC is unsuitable for the $tWSe_2$ model we briefly described in Sec. 1.2.2. Fortunately for us, we can show

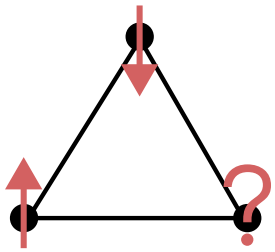


FIG. 2. Geometrical frustration in the triangular lattice of spins. Two of the spins are anti-aligned; the third is said to be frustrated, because it cannot simultaneously minimize interactions with both neighbours. The ground state is therefore degenerate.

that the ground state of certain systems is not just any generic many-body state picked at random from $\mathcal{H}$, but instead has a particular identifying feature which exponentially reduces the size of the parameter space our methods must search. The approach we will consider throughout the rest of this work takes advantage of this feature to scale linearly with N, works without modification in the thermodynamic limit, and generically works for a large class of Hamiltonians on any lattice: tensor network methods and the density matrix renormalization group (DMRG) algorithm.

1.4. Area law of entanglement entropy

As a preliminary, we investigate here the identifying structure of ground states. Consider a general pure many-body wavefunction $|\psi\rangle \in \mathcal{H}$ of a system of N particles confined to a lattice, with a single-particle Hilbert dimension d. For each bipartition of the system into parts A and B (see Fig. 3), we can associate an entanglement entropy between the partitions [37]. Consider, for example, examining partition B; the state of A is then given by the partial trace of the density matrix $\rho_{AB} = |\psi\rangle\langle\psi|$ over the Hilbert space of B,

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) \tag{2}$$

where we note that, in fact, $\rho_A = \rho_B := \rho$. The uncertainty in the state of A is then measured by the von Neumann entropy, known in the bipartite case as the entanglement entropy

$$S_{A|B}(\rho) = -\,\mathrm{Tr}[\rho \log(\rho)]. \tag{3}$$

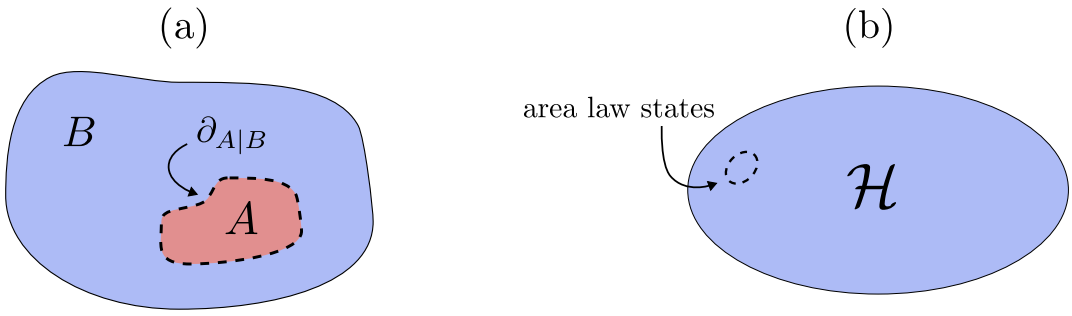


FIG. 3. (a) Bipartition of a system into a partition A and partition B; the dashed line indicates the boundary $\partial_{A|B}$ between them, which forms the basis of the area law. (b) The area law states constitute a tiny corner of the full many-body Hilbert space $\mathcal{H}$. Reproduced from [2].

Bipartite pure states have an entanglement which can be described entirely by the singular value decomposition (SVD), also known as the Schmidt decomposition [38] in the context of quantum physics (we will discuss this in more depth in Sec. 2; for now, a qualitative understanding will suffice). This leads to a so-called local unitary canonical form, in which the physically relevant nonlocal parameters (the Schmidt coefficients necessary for the characterization of correlations) and irrelevant local parameters are completely separated [39]. The bipartite entanglement thus has a simple quantitative structure which gives us insight into the corresponding physical reality. In principle, every measure of bipartite entanglement can be expressed through these Schmidt coefficients — entanglement entropy is perhaps the most well known of these measures [37].

We note that in considerations of multipartite entanglement, this is not, in general, the case; any number of different decompositions are possible, and they do not give us as deep of an insight into the underlying physical reality. The statements we make throughout this work follow the convention of condensed matter physics, in which *entanglement entropy* refers to bipartite entanglement (readers interested in the study of entanglement measures may find [39] helpful).

It has been shown that the entanglement entropy on pure states uniquely quantifies the degree of entanglement [40]. The entanglement entropy is a minimum $S_{A|B} = 0$ if A and B have no entanglement, i.e. form a product state $|\psi\rangle = |A\rangle \otimes |B\rangle$; it is bounded above by $S_{A|B} \leq |A| \log(d)$, where $|A|$ is the size (or volume) of the smallest partition [41]. A natural question then arises: can we, in general, classify the nature of states (such as ground states) by their entanglement entropy?

Fortunately for us, the answer is yes for an important class of systems. A state $|\psi\rangle \in \mathcal{H}$ picked at random (where by *random* we mean that the state is picked uniformly by sampling the Haar measure [42]) will, with overwhelming probability, be maximally entangled with $S_{A|B} = |A| \log(d)$, up to a correction that is exponentially small in $|B|$ [41]. Such states are referred to as *volume law states*, since their entanglement entropy scales with the volume $|A|$.

What scaling might we then expect from ground states? Recall that in condensed matter physics we are concerned primarily with local Hamiltonians, where by *local* we mean that the Hamiltonian can be written as a sum of operators H_i,

$$H = \sum_i H_i$$

where each operator H_i acts on a finite number of neighbours [8]. Physically, this corresponds to the fact that on the energy scales relevant to condensed matter, the dominant electromagnetic force generally decays rapidly with distance in materials due to screening effects, and so is effectively local [43]. Intuition might then guide us to propose that ground states of local Hamiltonians feature the strongest entanglement between neighbouring particles due to effectively finite-range interactions. This would then imply that the entanglement entropy for ground states is dominated by correlations between particles at the bipartition boundary; therefore, the entanglement entropy would scale with the size of that boundary (or area), i.e. $S_{A|B} \propto |\partial_{A|B}|$ where $|\partial_{A|B}|$ is the size of the boundary. Such states could then be referred to as *area law states*.

An actively growing body of research is reaching the consensus that our intuition is correct for ground states of gapped systems. It has been proved that gapped Hamiltonians (meaning here that there exists a gap in the eigenvalue spectrum of H between the ground state and the first excited state which persists in the thermodynamic limit $N \to \infty$) on arbitrary harmonic lattices indeed have ground state correlations which decay exponentially [44]. Similar exponential decay has been proved for gapped Hamiltonians on lattices which are translationally invariant in one direction [45]. The general conjecture that gapped Hamiltonians imply exponentially decaying correlations in the ground state is common wisdom within the field of quantum many-body physics, with Hastings and Koma of the highly influential [45] calling it

"perhaps... the most popular folk theorem in this field".

Further, Hastings has proved that for the ground states of 1d systems with gapped local Hamiltonians, the area law holds [46]; more precise bounds on the entanglement entropy have since been established [47]. In higher dimensions, it has been proved that the ground states of free gapped fermionic Hamiltonians satisfy the area law [44]. The evidence, we hope you'll agree, overwhelmingly indicates that ground states of gapped Hamiltonians are area law states (this is the consensus in the field of quantum many-body physics [48]).

The theory we have laid out thus far opens a potential route to circumventing the exponential scaling of the many-body Hilbert space: if we can construct the family of area law states, which form a tiny [41] corner of the overall Hilbert space (see Fig. 3), then finding the ground state through some variational minimization procedure becomes a viable strategy. In Sec. 2, we will employ tensor networks to construct and efficiently optimize precisely such states.

2. Tensor networks

In the notation of second quantization, a general, pure, many-body wavefunction $|\psi\rangle \in \mathcal{H}$ of a system of N particles is written as a superposition of single-particle basis states $\{|s_1\rangle, |s_2\rangle, ..., |s_N\rangle\}$,

$$|\psi\rangle = \sum_{s_1,...,s_N} \psi_{s_1,...,s_N} |s_1, ..., s_N\rangle \tag{4}$$

where $\psi_{s_1,...,s_N}$ is a rank-N tensor of complex coefficients with dimension $\dim(\psi_{s_1,...,s_N}) = d^N$, and d is the dimension of the single-particle Hilbert spaces. Therefore, the state is completely specified by knowledge of $\psi_{s_1,...,s_N}$. The idea of tensor networks is to take this big tensor and decompose it into a network of smaller tensors. By constructing tensor networks with specific structures, we can target much smaller subsets of the many-body Hilbert space which contain especially relevant states, such as the area law states.

2.1. Tensor diagram notation

Tensor network computations can quickly become unwieldy, with too many indices to keep track of. For this reason, tensor diagram notation has been developed as an elegant (but still rigorous) diagrammatic notation for tensor networks. The grammatical rules are simple:

- tensors are notated by shapes;
- tensor indices are notated by lines coming out of these shapes;
- lines which are connected imply a contracton over those indices.

The notation is intuitive enough that we will here consider it sufficient to provide a number of examples of tensor operations and objects in tensor diagram notation, and then explain anything else on a need-to-know basis. For a full introduction to tensor network diagrams, see, for example, [1]. Following directly from the rules we have set, we can easily write down diagrams representing arbitrary-rank tensors:

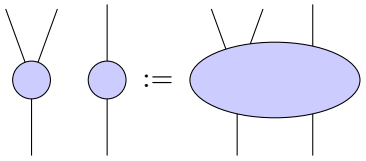
$x =$  (scalar)

$v_i =$  (vector)

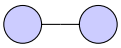
$M_{ij} =$  (matrix)

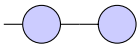
$A_{ijk} =$  (rank-3 tensor)

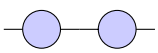
and so on. Performing operations with tensor diagram notation is similarly straightforward; as a particularly relevant example, tensor network methods make extensive use of the tensor product and contraction operations. The tensor product is formed by placing tensors next to one another, such that, for example, $A \otimes B = A_{ijk}B_{xy}$ is given by

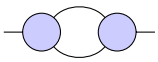
$A \otimes B :=$  .

Common contractions in the diagrammatic notation are given by, for example,

$\sum_i v_i u_i :=$  (vector dot product)

$\sum_j M_{ij} v_j :=$  (matrix-vector multiplication)

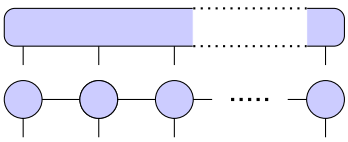
$\sum_j M_{ij} N_{jk} :=$  (matrix-matrix multiplication)

$\sum_{j,k} A_{ijk} B_{jkl} :=$  (two-index contraction of rank-3 tensors)

where readers familiar with the Einstein summation convention will notice that its implicit summation naturally arises within the diagrammatic notation, while having the distinct advantage of avoiding endless index management.

2.2. Matrix product states (MPS)

Equipped with tensor diagram notation, we can now easily see that there exists a representation of the coefficient tensor $\psi_{s_1,\ldots,s_N} := \psi$ of Eq. 4 as a network of N contracted tensors,

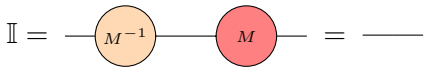
$\psi =$ 

$=$. (5)

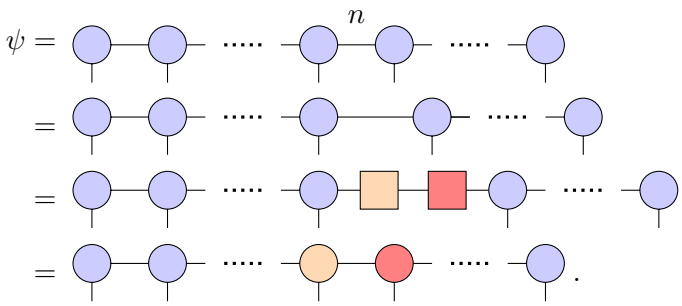
This is the *matrix product state* (MPS) decomposition of $|\psi\rangle$ [49, 50]. We have one tensor for each site on the lattice (or, alternatively, for each single-particle Hilbert space). Each tensor has a set of physical indices (the unconnected lines, where we note that for each tensor we have grouped the indices into one line corresponding to the local Hilbert space of dimension d for notational simplicity) and two sets of bond indices (note that at the edges we have only one set of bond indices, with the interpretation that the system has open boundary conditions; for periodic boundaries, tensor 1 and tensor N would have a bond).

2.3. Canonical form of the MPS

Motivated by Sec. 1.4, we now seek to relate this MPS decomposition to the entanglement entropy of the system. The MPS has given us access to a new parameter, the bond dimension m, which we conjecture may encode information about the entanglement between particles; we reason this by our earlier discussion about the existence of a local unitary canonical form of bipartite pure states, in which physically relevant nonlocal parameters and irrelevant local ones are completely separated. To this end, notice that the MPS is not unique, in the sense that it has a gauge freedom. Introduce the identity $M^{-1}M = \mathbb{I}$,



where in the tensor diagram notation the natural interpretation is that the identity lengthens a line, which has no overall effect on the network. Consider placing this identity on bond index n, and then performing the contraction of the left-hand tensor with M^{-1} (orange), and the right-hand tensor with M (red),



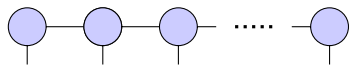
Under such gauge transformations of the bond indices, the physical state being described remains invariant — its representation, on the other hand, does not, i.e. gauge transformations of the MPS correspond to a change of bond basis. The *canonical gauge transformation* is the transformation which brings the state into the form of the Schmidt decomposition. Consider first the general many-body wavefunction of Eq. 4, and introduce a partition at site n into subsystems A and B. The Schmidt decomposition of this wavefunction is then

$$|\psi\rangle = \sum_{\alpha} \lambda_{\alpha}^{[n]} |A_\alpha\rangle_{s_1,\ldots,s_n} |B_\alpha\rangle_{s_{n+1},\ldots,s_N} \tag{6}$$

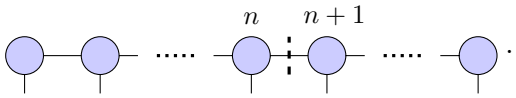
where $\lambda_\alpha^{[n]}$ are the Schmidt coefficients for a bipartition centered at n, and $|A_\alpha\rangle_{s_1,...,s_n}$, $|B_\alpha\rangle_{s_{n+1},...,s_N}$ are orthonormal basis states for A and B respectively. Such a decomposition is obtained by performing a singular value decomposition (SVD) on the tensor $\psi_{s_1,...,s_N}$ by transforming it into the form of a matrix. Defining collective indices $a = s_1, ..., s_n$, $b = s_{n+1}, ..., s_N$, the SVD tells us that there exists complex unitary matrices U, V and positive semi-definite diagonal matrix s (the namesake *singular values*, which by convention are sorted by magnitude) such that

$$\psi_{a,b} = UsV^\dagger. \tag{7}$$

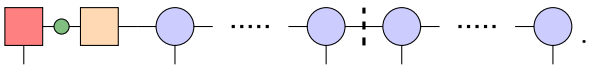
In the SVD form, U corresponds to $|A_\alpha\rangle_{s_1,...,s_n}$, s to λ_α, and $V^\dagger$ to $|B_\alpha\rangle_{s_{n+1},...,s_N}$. We wish to take advantage of the bond dimension parameter m made available by the MPS form of Eq. 5, and so we translate this SVD procedure to the language of tensor networks to determine the canonical form of the MPS. Consider Eq. 5,

$\psi =$ 

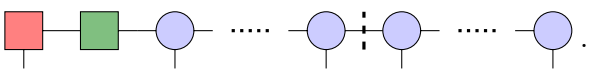
and introduce a partition on the bond index between tensors n and $n+1$,

$\psi =$ 

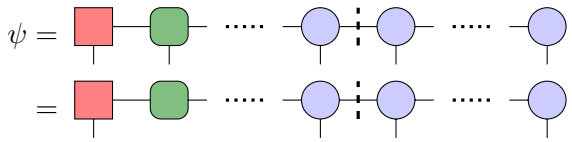
Consider the left-most tensor, and perform an SVD on it, denoting U by a red square, s by a green circle, and $V^\dagger$ by an orange square,

$\psi =$ 

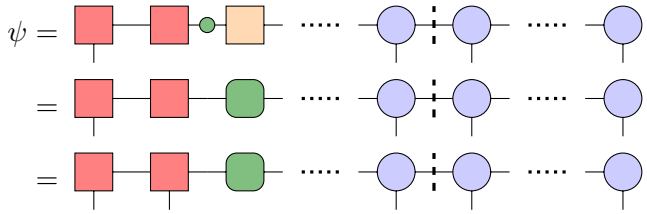
Now, contract the singular values s into $V^\dagger$ to form a matrix M_L (green square),

$\psi =$ 

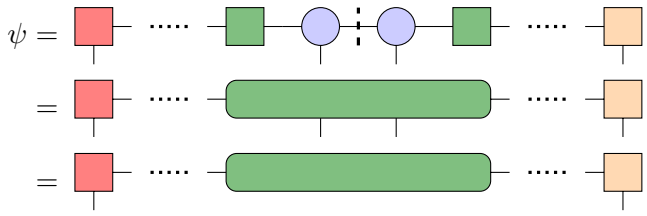
We now want to iteratively move M_L through the network, transforming each tensor in our path to the unitary form of U, until we reach the bond connecting tensor n and $n-1$. To move one bond up, we contract M_L with the tensor on its right, with the result a rank-3 tensor M'_L with a physical index (denoted by a green rounded square),



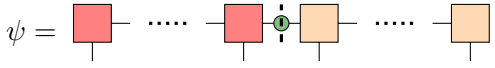
where in the second step we have grouped the physical index and the left-hand bond index such that M'_L has the form of a matrix. This allows us to perform an SVD on M'_L,



where in the final step we have ungrouped the physical index and the bond index on the left. This process can be continued for each tensor in the subsystem A. An identical process can then be done from the right-most tensor with contractions of the singular values s with U at each step, as opposed to $V^\dagger$, such that the matrices M_R travel backwards towards the partition boundary. We are then left with,



and applying one final SVD,



(8)

we have the local unitary canonical form (the Schmidt decomposition), given by the canonical gauge transformation. The interpretation of Eq. 8 is as follows: the tensors $U^{[i]}$, $V^{[j]\dagger}$ ($i \in \{1, ..., n\}$, $j \in \{n+1, ..., N\}$) are unitary by construction through the SVD,

$$U^{[i]}U^{[i]\dagger} = \mathbb{I},$$
$$V^{[j]\dagger}V^{[j]} = \mathbb{I}$$

and so we can identify the orthonormal basis states

$$|A\rangle = U^{[i]}|s_1, ..., s_n\rangle,$$
$$|B\rangle = V^{[j]\dagger}|s_{n+1}, ..., s_N\rangle.$$

The singular values $s^{[n]}$ therefore correspond to the Schmidt coefficients $\lambda^{[n]}$. In the canonical form, we can immediately read off the reduced density matrices in terms of the singular values,

$$\rho_A = \sum_\alpha \lambda_\alpha |A_\alpha\rangle|B_\alpha\rangle$$

and so directly calculate the entanglement entropy (recalling $\rho_A = \rho_B := \rho$),

$$\begin{aligned} S_{A|B} &= -\operatorname{Tr}[\rho \log(\rho)] \\ &= -\sum_\alpha \lambda_\alpha^2 \log(\lambda_\alpha^2) \end{aligned} \tag{9}$$

which we now understand to be entirely described by the singular values of the SVD. Still, it is not yet obvious that this form is any less numerically inaccessible than the original many-body wavefunction, since we are still describing the full many-body Hilbert space. Indeed, the canonical transformation carried out has the property that the bond dimension increases exponentially as one moves from the edges towards the partition [8], such that the bond dimensions for a system of N lattice sites looks like $(d, d^2, d^3, ..., d^N, ..., d^3, d^2, d)$ (for N even; for N odd, the maximum bond dimension is d^{N-1}).

Consider, however, fixing the bond dimension to some value $m \ll \exp(N)$ and enforcing a normalization

of the Schmidt coefficients $\lambda_\alpha = 1/\sqrt{m}$. It is straightforward to see then that the entanglement entropy is bounded above by a constant,

$$\begin{aligned} S_{A|B} &= -\sum^{m} \frac{1}{m} \log(\frac{1}{m}) \\ &= \sum^{m} \frac{1}{m} \log(m) \\ &= \log(m) \end{aligned}$$

i.e. the fixed-bond-dimension MPS obeys the one-dimensional area law (notice that the boundary size in 1d is constant in N, and so the area law is simply $S_{A|B} \propto c$ for some constant c). Thus, by fixing the bond dimension to some m, we restrict $\dim(\psi) = Ndm^2$, a monumental reduction of the many-body Hilbert space from exponential to linear growth in N. The states captured by the fixed-bond-dimension MPS are precisely the area law states which we have established to describe the ground states of local, gapped Hamiltonians. Thus, by restricting our simulations to the MPS family of fixed bond dimension, it becomes numerically feasible to search for the ground state of such models for much better aproximations of the thermodynamic limit $N \to \infty$.

Notice that fixing the bond dimension is equivalent to discarding all but the m largest Schmidt coefficients, since as we recall the singular values of the SVD are conventionally ordered by magnitude. This truncation makes good physical sense — we have established that most of the entanglement contributions (Schmidt coefficients) will be extremely small, and so we can significantly reduce the bond dimension, while incurring remarkably small errors, by discarding the majority of these entanglement contributions. Even for $m = \mathcal{O}(10)$, the fixed-bond-dimension MPS captures the entanglement properties of many systems quite well. In fact, it can be shown that this sort of truncation yields the optimal rank-m approximation with respect to the Frobenius norm [8].

2.4. The DMRG algorithm

We are now equipped with the set of area law states, which we have shown correspond to the fixed-bond-dimension MPS family. We now wish to perform some sort of minimization algorithm in order to converge on the ground state of a given local, gapped Hamiltonian. By far the most well understood and mature method in the study of quantum ground states by tensor networks (particularly in 1d) is the *density matrix renormalization group* (DMRG) algorithm, and so it is the method chosen for this work. Note that the name of the method holds little connection to the contemporary interpretation (the original formulation by White pre-dates the MPS, and is primarily physically motivated and dense [51]) and so we will exclusively refer to the method as DMRG moving forward.

A precise derivation of DMRG lies beyond the scope of this thesis (for the interested reader, an accessible introduction can be found in [1]; an in-depth contemporary review which focuses on the MPS formulation can be found in [49]). Instead, we will offer a qualitative explanation of the basis of DMRG in order to motivate its use here.

DMRG is rooted in the variational principle, which states that for any Hamiltonian H, suitably normalized wavefunction $|\psi\rangle$, and Rayleigh quotient

$$\varepsilon = \frac{\langle \psi \,|\, H \,|\, \psi \rangle}{\langle \psi | \psi \rangle}$$

we can make the following two statements: (1) $\varepsilon \geq H_0$, where H_0 is the ground state, and (2) $\varepsilon = H_0$ if and only if $|\psi\rangle$ is the wavefunction of the ground state. It is then implied that the ground state $|\psi_0\rangle$ can be found by performing a minimization of ε over some parameter space $\mathcal{P}$,

$$|\psi_0\rangle := \underset{|\psi\rangle \in \mathcal{P}}{\operatorname{argmin}} \left(\frac{\langle \psi \,|\, H \,|\, \psi \rangle}{\langle \psi | \psi \rangle} \right)$$

where in the case of DMRG we will take $\mathcal{P}$ to be the set of fixed-bond-dimension MPS state. The idea of DMRG, then, is to begin in some MPS state (a random MPS is generally a sufficient state for initialization, since it shares certain properties with ground states, but any educated estimate can be used), and then

variationally traverse $\mathcal{P}$, minimizing the Rayleigh quotient (energy) in each step.

The simplest algorithm in the DMRG family is the one-site DMRG; in each step, we pick a tensor, and then fix all other tensors. This tensor is then minimized, where the problem is simplified by gauge-fixing in the canonical form such that we have a transformation $H \to H'$. The resulting simplified optimization problem over H' then has a closed-form solution as the minimum eigenvectors of H'. The algorithm sweeps throughout the network, canonicalising each tensor and solving the simplified eigenvector problem.

The more advanced two-site DMRG blocks two tensors together at a time, performs a similar minimization routine as in one-site DMRG, and then splits the tensors back apart — the advantage is that this splitting process allows for dynamically controlling the bond dimension, and provides information about the truncation error in each step.

A number of high-performance libraries for research computation with tensor networks, including DMRG, are available; in this work, we use the ITensor (Julia, C++) library [52, 53], which implements a two-site DMRG method. We now define and motivate precisely the model of $tWSe_2$ homobilayers that we wish to explore.

3. Moiré Hubbard model

In recent years, a large body of research has focused on developing effective models of $tWSe_2$ homobilayers, which were initially identified as candidates for the study of strong correlation physics due to their strong interlayer coupling compared to heterobilayers [3]. In TMDs in general, the topmost valence bands originate from valleys at high-symmetry points in the Brillouin zone; *valleys* are local extrema in the band structure, such as the clear valleys in the peaks of Fig. 4 (c). For small-angle $tWSe_2$, scanning tunneling microscopy (STM) experiments [54] have found that the topmost bands originate from the $\pm K$ high symmetry points (where K conventionally labels the corners of the hexagonal Brillouin zone, see Fig. 4 (b)), and so initial theoretical discussions focused on these $\pm K$ valley states.

The spins are largely split between the $\pm K$ valence bands due to the strong spin-orbit coupling allowed

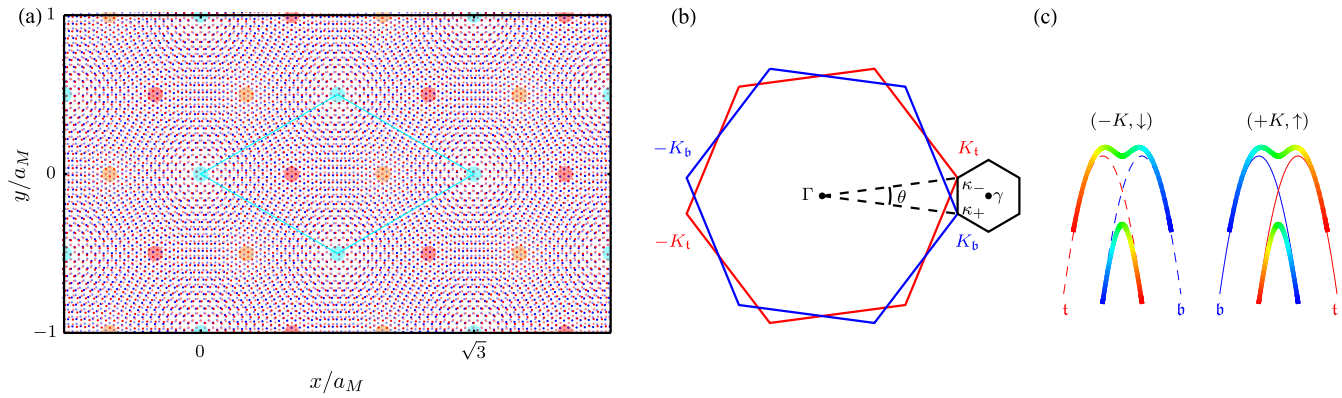


FIG. 4. (a) The formation of a moiré superlattice with lattice constant a_M when hexagonal WSe_2 lattices are stacked and twisted at small angles; cyan lines form the moiré unit cell. (b) The Brillouin zones associated with the top (red) and bottom (blue) layers, and the moiré Brillouin zone (black). (c) Schematic of the moiré band structure. All figures reproduced from [3].

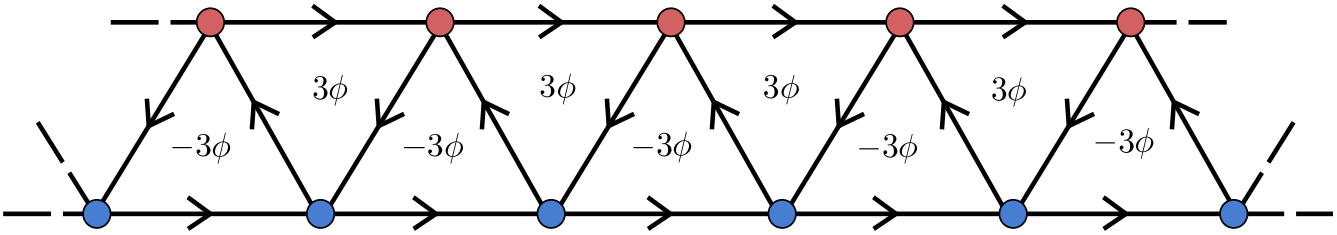


FIG. 5. Schematic of the moiré Hubbard model on a quasi-1d triangular ladder. Pairs of top (red) and bottom (blue) sites form the unit cell. Notice that this is the structure of the moiré superlattice in Fig. 4 (a).

by broken inversion symmetry in TMDs [55], which leads to an effective *spin-valley locking* in which the valley physics and spin physics become intertwined. This spin-valley locking reduces the number of symmetries present in the low-energy theory, by combining the spin and valley symmetries.

This reduction of the number of degrees of freedom is signficant, because it indicates that $tWSe_2$ is amenable to a relatively simple Hubbard description. This is part of the reason we chose to study TMD bilayers over graphene bilayers in Sec. 1.2.2, as graphene features spin, valley, and inversion symmetry [56]. Note, however, that the valleys are related by a time-reversal symmetry [57] which our Hubbard model will need to take into account.

3.1. Model of WSe_2 homobilayers

The Hubbard model which has developed in the field for the study of WSe_2 homobilayers is the so-called *moiré Hubbard model* [58], which differs from the standard Hubbard model by a spin-dependent phase in the hopping. In particular, we will consider an extended moiré Hubbard Hamiltonian

$$H = -t \sum_{\langle i,j \rangle} \left(c^\dagger_{i\uparrow} c_{j\uparrow} \exp(i\phi) + c^\dagger_{i\downarrow} c_{j\downarrow} \exp(-i\phi) + \text{h.c.} \right) + U_0 \sum_i n_{i\uparrow} n_{i\downarrow} + U_1 \sum_{\langle i,j \rangle} n_i n_j \tag{10}$$

on a quasi-1d triangular ladder (see Fig. 5), where t is the hopping constant between lattice sites i and j, ϕ is a spin-dependent phase factor, $c_{i\sigma}$ ($c^\dagger_{i\sigma}$) is the fermionic annihilation (creation) operator for a spin-σ electron at site i, U_0, U_1 are the on-site and nearest-neighbour density interaction strengths respectively, $n_{i\sigma} = c^\dagger_{i\sigma} c_{i\sigma}$ is the occupation number operator which counts the spin$-\sigma$ electrons at site i, and $n_i = c^\dagger_{i\uparrow} c_{i\uparrow} + c^\dagger_{i\downarrow} c_{i\downarrow}$ is the total occupation number operator which counts all electrons at site i. The index $\langle i, j \rangle$ on the summation means the nearest neighbours j to site i.

The unique component of the moiré Hubbard model is the spin-dependent phase on the hopping constant $t \exp(i\sigma\phi)$, initially obtained from density functional theory calculations [58], and which corresponds to the displacement field between the layers. This phase is the simplest way of breaking inversion symmetry, and its spin-dependence ensures time-reversal symmetry. The displacement field acts perpendicular to the 2d sheet of WSe_2, and so through each triangle in the lattice we have a total flux $3\sigma\phi$, which alternates in sign between neighbouring triangles. As the electrons move clockwise around a triangle on the lattice (note the arrows in Fig. 5), they gain a $+\phi$ phase for each hop; as they move counter-clockwise, they gain a $-\phi$ phase. Note that the model is invariant under a transformation $\phi \to \phi + \frac{2\pi}{3}$, since then we would have a flux per triangle

$$\left| \pm 3 \left(\phi + \frac{2\pi}{3} \right) \right| = 3\phi + 2\pi = 3\phi$$

where the flux is 2π periodic. Thus, the physics of the model is completely described by the range $[\phi, \phi + \frac{2\pi}{3}]$. Experimentally, the electron density and ϕ are highly tunable: the sum of the layer voltages determines the chemical potential (which in turn determines the electron density), and their difference

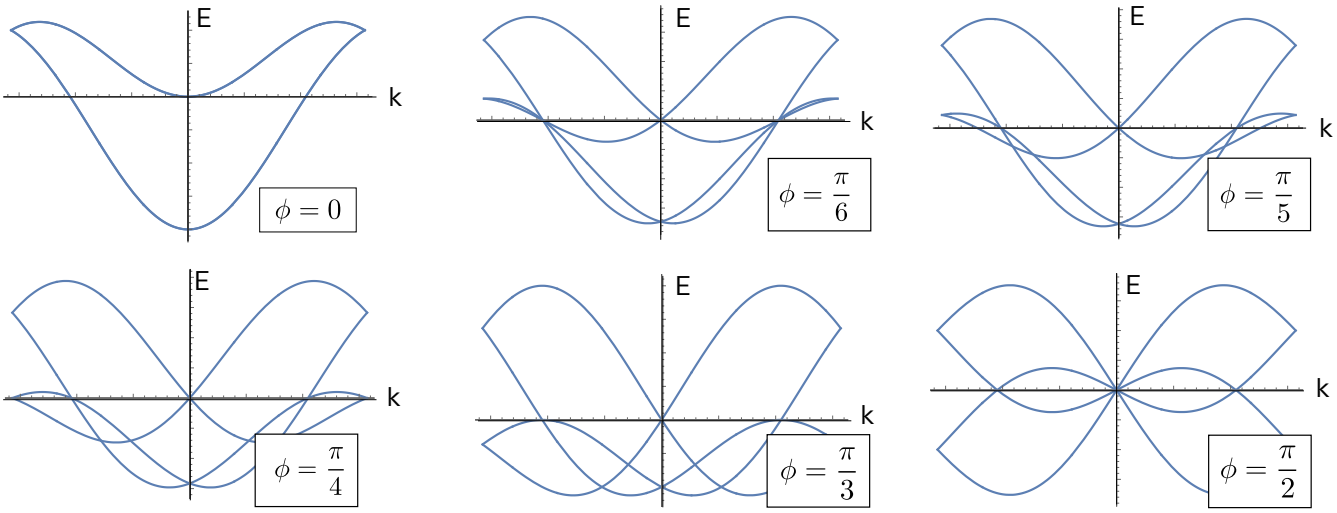


FIG. 6. The band structure of our model in the non-interacting limit, plotted from the energy-momentum dispersion relations $E(k)$ (obtained from the diagonalization of Eq. 12) for a variety of values of $\phi < \frac{2\pi}{3}$.

determines ϕ. Displacement fields in the approximate range of $-\frac{\pi}{3} \lesssim \phi \lesssim \frac{\pi}{3}$ are experimentally achievable [58]. We therefore will focus on strongly correlated physics which is controlled by these two experimental degrees of freedom.

3.2. Non-interacting limit

A reasonable starting point to our analysis is to consider the model in the non-interacting limit, U_0, $U_1 = 0$, where the Hamiltonian reads

$$H = -t \sum_{\langle i,j \rangle} \left(c^\dagger_{i\uparrow} c_{j\uparrow} \exp(i\phi) + c^\dagger_{i\downarrow} c_{j\downarrow} \exp(-i\phi) + \text{h.c.} \right).$$

This is a tight-binding Hamiltonian, and so we can diagonalize it to yield the band structure. To accomplish this, note that the lattice can be decomposed into a top lattice (red) and a bottom lattice (blue). The unit cell is then a pair of one top and one bottom site. By indexing the unit cells by j, we can write

the sum over nearest neighbours explicitly,

$$\begin{aligned} H = -t \sum_j \big[& c^\dagger_{jb\uparrow} c_{jt\uparrow} \exp(i\phi) + c^\dagger_{jt\uparrow} c_{(j+1)b\uparrow} \exp(i\phi) + c^\dagger_{jb\uparrow} c_{(j+1)b\uparrow} \exp(-i\phi) \\ & + c^\dagger_{jt\uparrow} c_{(j+1)t\uparrow} \exp(-i\phi) + c^\dagger_{jb\downarrow} c_{jt\downarrow} \exp(-i\phi) + c^\dagger_{jt\downarrow} c_{(j+1)b\downarrow} \exp(-i\phi) \\ & + c^\dagger_{jb\downarrow} c_{(j+1)b\downarrow} \exp(i\phi) + c^\dagger_{jt\downarrow} c_{(j+1)t\downarrow} \exp(i\phi) + \text{h.c.} \big] \end{aligned}$$

where we have defined fermionic annihilation (creation) operators $c_{js\sigma}$ ($c^\dagger_{js\sigma}$) for the spin-σ electron at the $s =$ (top or bottom) cell in unit cell j. For simplicity, write $H = H_\uparrow + H_\downarrow$, and consider for now only $H_\uparrow$,

$$\begin{aligned} H_\uparrow = -t \sum_j \big[& c^\dagger_{jb\uparrow} c_{jt\uparrow} \exp(i\phi) + c^\dagger_{jt\uparrow} c_{(j+1)b\uparrow} \exp(i\phi) + c^\dagger_{jb\uparrow} c_{(j+1)b\uparrow} \exp(-i\phi) \\ & + c^\dagger_{jt\uparrow} c_{(j+1)t\uparrow} \exp(-i\phi) + \text{h.c.} \big]. \end{aligned} \tag{11}$$

Assume periodic boundary conditions such that we have translational symmetry along the x direction. Then, we can define a canonical transformation of the operators to momentum space by applying the discrete Fourier transform in 1d,

$$\begin{aligned} c_{js\sigma} &= \frac{1}{\sqrt{L}} \sum_k \exp(ikj) c_{ks\sigma} \\ c^\dagger_{js\sigma} &= \frac{1}{\sqrt{L}} \sum_k \exp(-ikj) c^\dagger_{ks\sigma} \end{aligned}$$

where $L = \frac{N}{2}$ is the number of unit cells, N is the number of sites, and k is the quasimomentum. Substituting into $H_\uparrow$ (Eq. 11),

$$\begin{aligned} H_\uparrow = -\frac{t}{L} \sum_j \sum_{k,k'} \big[& \exp(i\phi) c^\dagger_{kb\uparrow} c_{kt\uparrow} + \exp(ij[k'-k]) \exp(ik' + i\phi) c^\dagger_{kt\uparrow} c_{k'b\uparrow} \\ & + \exp(ij[k'-k]) \exp(ik' - i\phi) c^\dagger_{kb\uparrow} c_{k'b\uparrow} + \exp(ij[k'-k]) \exp(ik' - i\phi) c^\dagger_{kt\uparrow} c_{k't\uparrow} + \text{h.c.} \big] \end{aligned}$$

and recalling the definition of the dirac delta,

$$\delta_{k'k} = \frac{1}{L}\sum_j \exp(ij[k'-k])$$

we have

$$H_\uparrow = -t\sum_k \big[\exp(i\phi)c^\dagger_{kb\uparrow}c_{kt\uparrow} + \exp(ik+i\phi)c^\dagger_{kt\uparrow}c_{kb\uparrow} + \exp(ik-i\phi)c^\dagger_{kb\uparrow}c_{kb\uparrow} + \exp(ik-i\phi)c^\dagger_{kt\uparrow}c_{kt\uparrow} + \text{h.c.}\big].$$

Writing out the hermitian conjugate explicitly,

$$\begin{aligned}
H_\uparrow = -t\sum_k \big[\ & \exp(i\phi)c^\dagger_{kb\uparrow}c_{kt\uparrow} + \exp(-i\phi)c^\dagger_{kt\uparrow}c_{kb\uparrow} \\
& + \exp(ik+i\phi)c^\dagger_{kt\uparrow}c_{kb\uparrow} + \exp(-ik-i\phi)c^\dagger_{kb\uparrow}c_{kt\uparrow} \\
& + \exp(ik-i\phi)c^\dagger_{kb\uparrow}c_{kb\uparrow} + \exp(-ik+i\phi)c^\dagger_{kb\uparrow}c_{kb\uparrow} \\
& + \exp(ik-i\phi)c^\dagger_{kt\uparrow}c_{kt\uparrow} + \exp(-ik+i\phi)c^\dagger_{kt\uparrow}c_{kt\uparrow}. \\
= -t\sum_k \big[\ & \big(\exp(i\phi) + \exp(-ik-i\phi)\big)c^\dagger_{kb\uparrow}c_{kt\uparrow} \\
& + \big(\exp(-i\phi) + \exp(ik+i\phi)\big)c^\dagger_{kt\uparrow}c_{kb\uparrow} \\
& + 2\cos(k-\phi)c^\dagger_{kb\uparrow}c_{kb\uparrow} + 2\cos(k-\phi)c^\dagger_{kt\uparrow}c_{kt\uparrow}.
\end{aligned}$$

An identical procedure can be performed for $H_\downarrow$, yielding (in a form that is easier on the eyes),

$$\begin{aligned}
H_\uparrow &= \sum_k C^\dagger_{k\uparrow}h_\uparrow C_{k\uparrow}, \\
h_\uparrow &= -t\begin{bmatrix} 2\cos(k-\phi) & \exp(i\phi)+\exp(-ik-i\phi) \\ exp(-i\phi)+\exp(ik+i\phi) & 2\cos(k-\phi)\end{bmatrix} \qquad C_{k\uparrow} = -t\begin{bmatrix} c_{kb\uparrow} \\ c_{kt\uparrow}\end{bmatrix} \\
H_\downarrow &= \sum_k C^\dagger_{k\downarrow}h_\downarrow C_{k\downarrow}, \\
h_\downarrow &= -t\begin{bmatrix} 2\cos(k+\phi) & \exp(-i\phi)+\exp(-ik+i\phi) \\ exp(i\phi)+\exp(ik-i\phi) & 2\cos(k+\phi)\end{bmatrix} \qquad C_{k\downarrow} = -t\begin{bmatrix} c_{kb\downarrow} \\ c_{kt\downarrow}\end{bmatrix}.
\end{aligned}$$

Then, the full non-interacting Hamiltonian in momentum space is

$$H = H_\uparrow + H_\downarrow = \sum_k \left(C^\dagger_{k\uparrow} h_\uparrow C_{k\uparrow} + C^\dagger_{k\downarrow} h_\downarrow C_{k\downarrow} \right)$$

or in a form more amenable to numerical diagonalization,

$$\begin{aligned} H &= \sum_k C^\dagger_k h C_k, \\ h &= -t \begin{bmatrix} h_{\uparrow bb} & h_{\uparrow bt} & 0 & 0 \\ h_{\uparrow tb} & h_{\uparrow tt} & 0 & 0 \\ 0 & 0 & h_{\downarrow bb} & h_{\downarrow bt} \\ 0 & 0 & h_{\downarrow tb} & h_{\downarrow tt} \end{bmatrix} \qquad C_{k\uparrow} = -t \begin{bmatrix} c_{kb\uparrow} \\ c_{kt\uparrow} \\ c_{kb\downarrow} \\ c_{kt\downarrow} \end{bmatrix}. \end{aligned} \tag{12}$$

Standard numerical diagonalization of h then gives us the energy-momentum dispersion relations $E(k)$, which we plot in Fig. 6 for a number of values of ϕ. To discuss what this band structure is telling us, we introduce the notion of a *chemical potential* μ upon which the electron density $\rho(\mu) := \rho$ depends. For our purposes, it is sufficient to understand μ as the parameter in the Fermi fuction

$$F(k) = \frac{1}{\exp([E(k) - \mu]/k_B T) + 1}$$

which separates the filled states from the empty states. In the zero-temperature limit of the ground state, $T \to 0$ and $F(k)$ is constrained to be either 0 when $E(k) > \mu$, or 1 when $E(k) < \mu$, and in this way counts the number of states which are filled, which are the states below the chemical potential. We can visualize this directly on plots of the band structure by drawing a line at $E(k) = \mu$ (see Fig. 7 (a)). The energy of the states in the non-interacting limit is completely dependent on how many electrons are in the system, and so ρ increases as μ does.

Fig. 6 then has the interpretation that in the non-interacting limit, the system is completely metallic: as μ is increased through the bands, there are no gaps in the band structure at which the density of states is 0. This is intuitively the expected result, as the electrons are completely free to enter the system as soon as the chemical potential is sufficient. We are then motivated to pose the following question: *can*

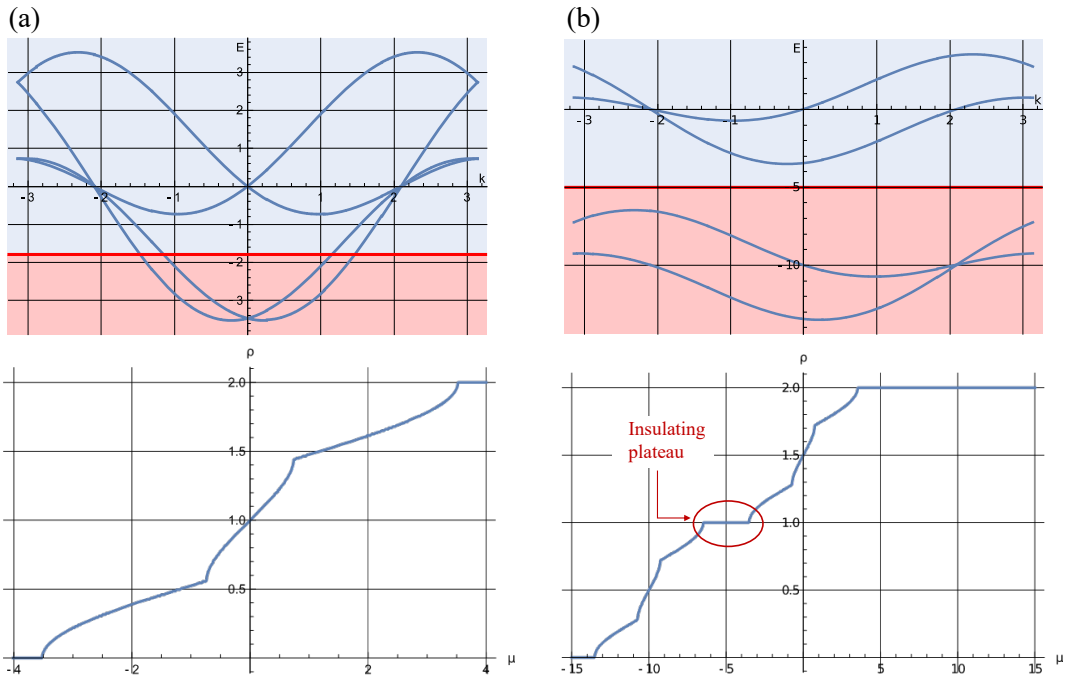


FIG. 7. (a) The band structure at $\phi = \frac{\pi}{6}$ and the corresponding electron density $\rho(\mu)$. The chemical potential μ is plotted as a red line; states below (shaded in red) are filled, and states above (shaded in blue) are empty. Note the kinks in ρ at $\mu \approx \pm 1$, which may correspond to Van Hove singularities. (b) An artificial example of a band gap; the corresponding plot of μ features an insulating plateau.

the presence of interactions at particularly sensitive areas of the band structure open a band gap to reveal exotic insulating states?

In the interacting case, we will not have direct access to the band structure, and so we cannot identify insulating states by the presence of a band gap. However, we know that in a band gap changes in μ will not lead to the filling of new states (i.e. the increase of the electron density), and so in a plot of $\rho(\mu)$ we will see so-called insulating plateaus developing (the compressibility $\frac{d\rho}{d\mu} = 0$). As an example, Fig. 7 (b) illustrates a band gap in an artifically designed non-interacting Hamiltonian, and the corresponding insulating plateau in the electron density. It is the latter by which we can identify insulating states in the interacting case.

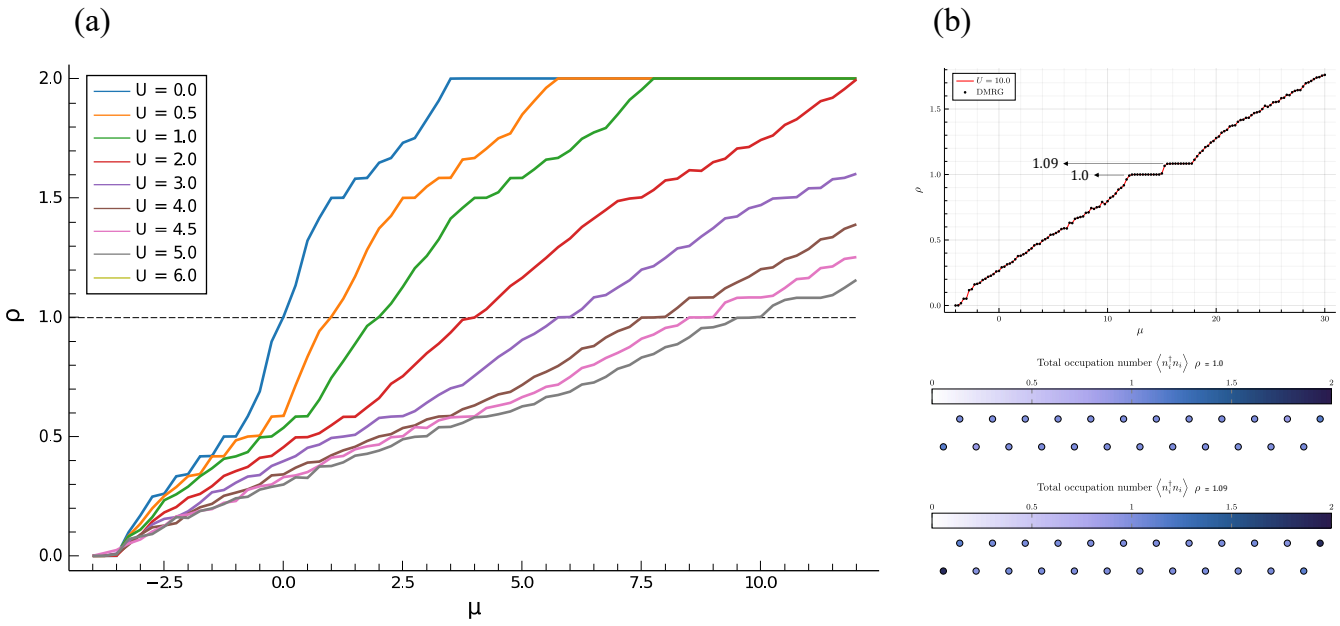


FIG. 8. (a) The development of a Mott insulating state at $\frac{1}{2}$ filling ($\rho = 1$) as the interaction strength U increases on an $L = 12$ ladder. (b) A well-developed Mott insulating state on the $L = 12$ ladder with $U = 10$. Note the anomalous plateau at $\rho \approx 1.09$. Below, the expectation value of the total occupation number $\langle n_i^\dagger | n_i \rangle$ is plotted for both plateaus.

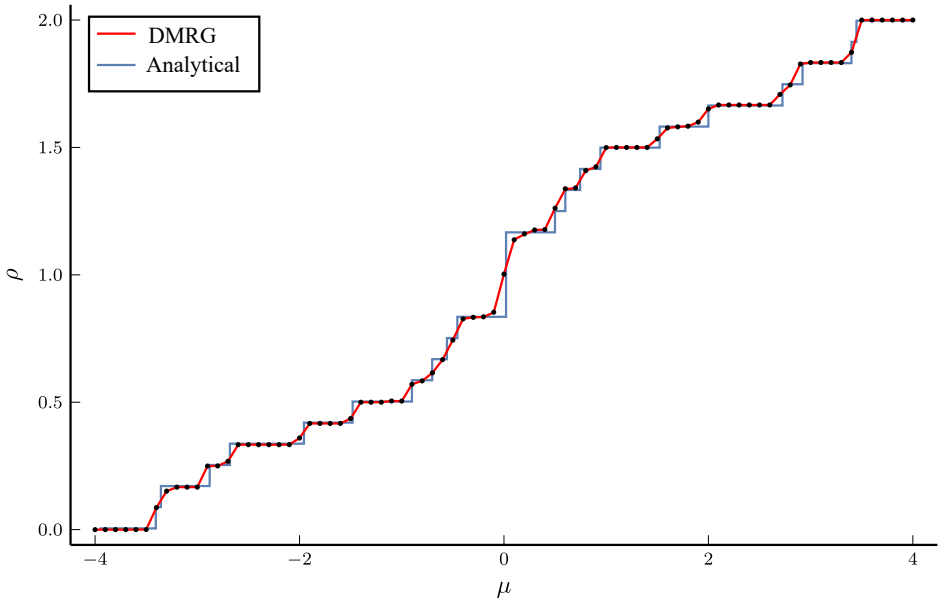


FIG. 9. Comparison of the DMRG-calculated electron density ρ of the non-interacting ground states to the analytical solution. Even with a relatively small final bond dimension of $m = 200$, we note good qualitative agreement.

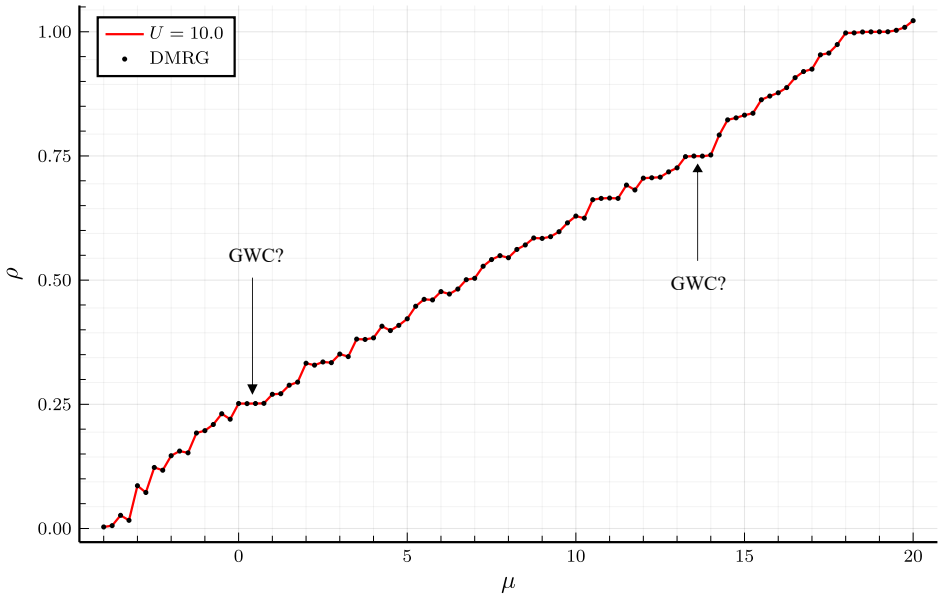


FIG. 10. Potential insulating plateaus at $\frac{1}{8}$ and $\frac{3}{8}$ fillings, which may correspond to generalised Wigner crystal (GWC) states.

4. DMRG results in the interacting model

For the DMRG, we note that our model is quasi-1d, in that it may be mapped to a 1d chain with on-site, nearest neighbour, and next nearest neighbour interactions by translating one of the sublattices along the y direction such that the sublattices coincide. The DMRG is then optimal. Note, further, that all DMRG calculations were done with open boundary conditions. DMRG, in general, optimizes an open boundary MPS, and so while DMRG can handle periodic boundary conditions, it converges on the ground state significantly slower. All DMRG calculations were performed for $L = 12$ unit cell systems (24 sites total) and were truncated at a bond dimension of $m = 200$ unless otherwise specified. For simplicity, we combine the interaction strengths by defining $U_0 := U$, $U_1 := \alpha U$, where α is a scaling factor that determines the relative strengths. For all simulations presented, $\alpha = 0.25$ unless stated otherwise.

Notice that, of the band structures in Fig. 6, three bands for $\frac{\pi}{6}$ (the most extreme possible phase factor due to the periodicity) intersect — we conjecture that interactions at this point may open a band gap. We first perform a benchmark of the DMRG against the known analytical electron density $\rho(\mu)$ (Fig. 9) and note good qualitative agreement even for a severe bond dimensions truncation at $m = 200$. Differences

to the analytical are primarily due to a lack of resolution in μ.

In Fig. 8 (a), we plot ρ for different interaction strengths U. We find that at $U \approx 2$ an insulating plateau begins to form at $\frac{1}{2}$ filling ($\rho = 1$); by $U \approx 3$ it has fully developed. As U increases, the length of this plateau increases but remains stable. This is the Mott insulating state at $\frac{1}{2}$ filling which is driven by strong electron correlations. In (b), we investigate the plateau at high interaction strength $U = 10$ with more fidelity (maximum bond dimension truncation at $m = 400$, and a higher density of calculations). The Mott state is pronounced, and plotting the expectation value of the total occupation number operator $\langle n_i^\dagger | n_i \rangle$ indeed confirms that there is one electron per site. We also, however, find a pronounced anomalous insulating plateau just above the expected Mott insulator, at a lattice filling of $\frac{13}{24}$. In fact, examining (a) we can see this plateau also developing at $U \approx 3$ and growing alongside the Mott state. Examining the occupation, we can see that the ground state is identical to the Mott state, except for the presence of two electrons in the corners of the ladder — we thus believe this to be a non-physical finite-size effect caused by the open boundary conditions. We expect that in the thermodynamic limit $L \to \infty$ this plateau would vanish.

Away from $\frac{1}{2}$ filling, and with a a stronger nearest-neighbour interaction scaling of $\alpha = 0.4$, we find inconclusive evidence (see Fig. 10) of generalised Wigner crystal (GWC) states at $\frac{1}{8}$ filling ($\rho = 0.25$) and $\frac{3}{8}$ filling ($\rho = 0.75$). If we assume the electrons localize at the sites such that they have no neighbouring electrons, the $\frac{1}{8}$ filling GWC would be the least dense GWC possible on the $L = 12$ ladder. Unfortunately, due to limited computational resources, these states were not pursued further here.

5. Conclusion and discussion

The high tunability of WSe_2 homobilayers (particularly the freedom to vary the electron density and hopping phase) makes the moiré material an ideal candidate for the comparison of Hubbard model physics to experiment. We have shown here that a fairly simple analysis of the moiré Hubbard model already reveals an exotic Mott insulating state, and indications of generalised Wigner crystal states. We have also, in doing this analysis, demonstrated the power of tensor network methods, and DMRG in particular,

in uncovering the complex inner workings of strongly correlated materials. In the age of MPS, tensor network methods may be key to understanding the collective quantum behaviour inherent in the many-body systems that form our world.

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